Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

IoT with Raspberry Pi

TGS Ref No: TGS-2020505433

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 15.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
14.0	20 February 2023	Previous release — 'Mastering Raspberry Pi: Hands-On Practical Applications for Beginners', 2-day structure with ThingSpeak, MATLAB, Node-RED, Adafruit IO and IFTTT practical activities.	Dr. Alfred Ang
15.0	8 August 2026	Major revision. Course retitled from 'Mastering Raspberry Pi: Hands-On Practical Applications for Beginners' to 'IoT with Raspberry Pi'. All practical activities replaced with four hands-on labs on the IoTFlow platform (https://iot.tertiaryinfotech.com); the legacy ThingSpeak, MATLAB, Node-RED, Adafruit IO and IFTTT lab set retired. Deck rebuilt to the current house visual standard. Assessment stated as WA (Q&A) 60 min plus Practical Performance 60 min.	Dr. Alfred Ang

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Course Information

Course Title	IoT with Raspberry Pi
WSQ Course Reference	TGS-2020505433
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
TSC Title / Code	Internet of Things Management · EVS-TEM-4002-1.1 (Level 4)
Duration	2 days · 16 training hours (excluding a 1-hour lunch break each day)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks counted within training time)
Course Components	Classroom Facilitation 9.0 h · Practical / Practicum 5.0 h · Assessment 2.0 h
Mode	Instructor-led classroom facilitation with hands-on IoT labs on the IoTFlow cloud platform
Platform	IoTFlow — https://iot.tertiaryinfotech.com
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Analyse the uses and functions of IoT technologies. (A1, K1)
- LO2: Collect environment data with Raspberry Pi and post the data to cloud computing. (A2, K2, K3)
- LO3: Use cloud computing to analyse and visualise IoT data. (A3, K4, K5)
- LO4: Monitor data and trigger control on devices. (A4, K6, K7)

Skills Framework Mapping (TSC EVS-TEM-4002-1.1 Internet of Things Management)

Manage the deployment of Internet of Things (IoT) solutions, including the collection and analysis of data from connected sensors and devices, to monitor and control equipment and the environment, and to drive process improvement.

Abilities (assessed by the Practical Performance)	Knowledge (assessed by the Written Assessment)
A1 — Analyse relevant data to recommend control actions	**K1 —** IoT concept and technical knowledge of IoT implementation
A2 — Identify areas for implementing IoT solutions for process improvement	**K2 —** Connectivity in devices using sensors, smart devices and other technologies for data collection and communication
A3 — Use relevant tool to analyse and record performance	**K3 —** Equipment automation
A4 — Monitor the effectiveness of IoT solutions	**K4 —** Building and/or Environment automation
	K5 — Advanced process control
	K6 — Security and privacy applications for IoT
	K7 — IoT guidelines and communication

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	standards

Assessment

- Written Assessment (WA) — Short-Answer Questions (Q&A), 60 minutes, open book. Covers K1–K7.
- Practical Performance (PP) — 60 minutes, open book, individually completed. Covers A1 –A4.
- Format: Open Book — course slides, Learner Guide and approved materials only.
- The assessment is conducted at the end of Day 2, from 5:10 pm.
- Assessor-to-learner ratio: 1:3 to 1:10 for both the WA (Q&A) and the Practical Performance.
- A minimum of 75% attendance is required to be eligible for assessment and funding.

Course Schedule

Day 1 — IoT Foundations and Collecting Data — From Sensor to Cloud

Time	Duration	Topic / Activity
9:30am–9:45am	15 min	Welcome, trainer and learner introductions, ground rules, and mandatory digital attendance (AM)
9:45am–10:00am	15 min	Learning outcomes, course outline and Skills Framework (TSC EVS-TEM-4002-1.1) briefing
10:00am–10:45am	45 min	**Topic 1 — Overview of Internet of Things (IoT), part 1: the 4th Industrial Revolution, what IoT is, the four building blocks of an IoT system, and how IoT works. (LO1 / A1, K1)**
10:45am–11:00am	15 min	Tea break
11:00am–11:55am	55 min	**Topic 1 — Overview of Internet of Things (IoT), part 2: sensors and actuators, wireless technologies for IoT, types of IoT and IIoT, and IoT use cases across industry. (LO1 / A1, K1)**
11:55am–12:30pm	35 min	**Topic 1 — Facilitated discussion 'Analyse an IoT Use Case' and topic recap. (LO1 / A1, K1)**
12:30pm–1:30pm	60 min	Lunch break
1:30pm–1:40pm	10 min	Mandatory digital attendance (PM)
1:40pm–2:35pm	55 min	**Topic 2 — Collect and Post Environmental Data to Cloud (concepts): cloud computing**

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		for IoT, the IoTFlow platform, the Raspberry Pi as an IoT gateway, environmental sensors, device tokens, and the HTTP REST and MQTT uplink. (LO2 / A2, K2, K3)
2:35pm–3:20pm	45 min	**Topic 2 — HANDS-ON PRACTICAL on IoTFlow (https://iot.tertiaryinfotech.com). Lab 1: Register Your Raspberry Pi on IoTFlow**
3:20pm–3:35pm	15 min	Tea break
3:35pm–5:35pm	120 min	**Topic 2 — HANDS-ON PRACTICAL (continued). Lab 2: Collect Environmental Data and Post It to the Cloud**
5:35pm–6:10pm	35 min	**Topic 2 — Reviewing the uplink: verifying telemetry arrival, troubleshooting connectivity, running the sensor script unattended, and topic recap. (LO2 / A2, K2, K3)**
6:10pm–6:30pm	20 min	Day 1 recap, Q&A, and preview of Day 2

Day 1 training time: 480 minutes (8 hours), excluding the 1-hour lunch break. Classroom Facilitation 315 min · Practical/Practicum 165 min.

Day 2 — Analytics, Visualisation and Control — From Data to Action

Time	Duration	Topic / Activity
9:30am–9:40am	10 min	Mandatory digital attendance (AM) and Day 1 recap
9:40am–10:35am	55 min	**Topic 3 — IoT Data Analytics and Visualization (concepts): why visualise, display widgets, analysing telemetry history for trends and drift, alert rules, and recording performance. (LO3 / A3, K4, K5)**
10:35am–10:50am	15 min	Tea break
10:50am–12:05pm	75 min	**Topic 3 — HANDS-ON PRACTICAL on IoTFlow. Lab 3: Analyse and Visualise Your IoT Data on the Cloud**
12:05pm–12:30pm	25 min	**Topic 3 — Facilitated discussion 'Design Your Monitoring Dashboard' and topic recap. (LO3 / A3, K4, K5)**
12:30pm–1:30pm	60 min	Lunch break
1:30pm–1:40pm	10 min	Mandatory digital attendance (PM)
1:40pm–2:35pm	55 min	**Topic 4 — Monitor Data to Trigger Control from Cloud (concepts): reading data back over REST and MQTT,**

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		downlink control and virtual pins, automation with n8n, monitoring effectiveness, and IoT security. (LO4 / A4, K6, K7)
2:35pm–3:35pm	60 min	**Topic 4 — HANDS-ON PRACTICAL on IoTFlow. Lab 4: Monitor Data and Trigger Control on the Raspberry Pi**
3:35pm–3:50pm	15 min	Tea break
3:50pm–4:15pm	25 min	**Topic 4 — Facilitated discussion 'Monitor Effectiveness and Manage Risk', IoT guidelines and communication standards, and topic recap. (LO4 / A4, K6, K7)**
4:15pm–4:30pm	15 min	Course summary and Q&A, TRAQOM survey, Assessment digital attendance, and Briefing for Assessment
4:30pm–5:30pm	60 min	**Written Assessment (WA) — Short-Answer Questions (Q&A), 60 minutes, open book. Assesses knowledge K1–K7.**
5:30pm–6:30pm	60 min	**Practical Performance (PP) — 60 minutes, open book, individually completed. Assesses abilities A1–A4.**

Day 2 training time: 480 minutes (8 hours), excluding the 1-hour lunch break. Classroom Facilitation 225 min · Practical/Practicum 135 min · Assessment 120 min.

Course total: 960 minutes (16 hours) across 2 training days, excluding lunch breaks. Classroom Facilitation 540 min (9.0 h) · Practical/Practicum 300 min (5.0 h) · Assessment 120 min (2.0 h).

Topic and Activity Reference

Topic / Learning Unit	Outcome & TSC mapping	In-class activity
Topic 01: Overview of Internet of Things (IoT)	LO1 · A1, K1 · Day 1 AM	Discussion — Analyse an IoT Use Case (20 min facilitated discussion)
Topic 02: Collect and Post Environmental Data to Cloud	LO2 · A2, K2, K3 · Day 1 PM	Lab 1: Register Your Raspberry Pi on IoTFlow, Lab 2: Collect Environmental Data and Post It to the Cloud
Topic 03: IoT Data Analytics and Visualization	LO3 · A3, K4, K5 · Day 2 AM	Lab 3: Analyse and Visualise Your IoT Data on the Cloud
Topic 04: Monitor Data to Trigger Control from Cloud	LO4 · A4, K6, K7 · Day 2 PM	Lab 4: Monitor Data and Trigger Control on the Raspberry Pi

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Hands-On Labs (Topics 2–4 — Practical / Practicum, 300 minutes)

All labs run on IoTFlow (https://iot.tertiaryinfotech.com). Full step-by-step instructions are in the Learner Guide; the slide deck presents each lab's purpose and outcome only.

Lab	Title	What the learner builds
Lab 1	Register Your Raspberry Pi on IoTFlow	A registered Raspberry Pi device in your own IoTFlow project, with its device token saved
Lab 2	Collect Environmental Data and Post It to the Cloud	Live temperature and humidity telemetry streaming from the Raspberry Pi into IoTFlow
Lab 3	Analyse and Visualise Your IoT Data on the Cloud	A live monitoring dashboard for your Raspberry Pi, plus a working threshold alert rule
Lab 4	Monitor Data and Trigger Control on the Raspberry Pi	A dashboard switch and slider that control an LED/relay on the Raspberry Pi in real time

Instructional Methods

- Lecture — trainer-led explanation of concepts with visual slides.
- Didactic questioning — the trainer poses questions to check and deepen understanding.
- Facilitated discussion — learners apply each topic to their own organisation (Topics 1, 3 and 4).
- Demonstration — the trainer demonstrates each IoTFlow workflow before the learners attempt it.
- Hands-on practical — learners complete Labs 1–4 on their own IoTFlow project (Topics 2–4).

Resources Required

- Classroom with projector or large screen, whiteboard and high-speed WiFi.
- One laptop per learner with a modern web browser (spare laptops are available).
- A free learner account on IoTFlow (https://iot.tertiaryinfotech.com).
- Python 3 with the IoTFlow client (pip install "iotflow[mqtt]") for the optional scripted labs.
- IoT hardware (ESP32/ESP8266) is optional — every lab can be completed from the browser and terminal.
- Course slides, Learner Guide and Lesson Plan available on https://lms-tms.tertiaryinfotech.com.

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